

Amyloid-β directed: 3 molecules stopped early, all monoclonal antibodies

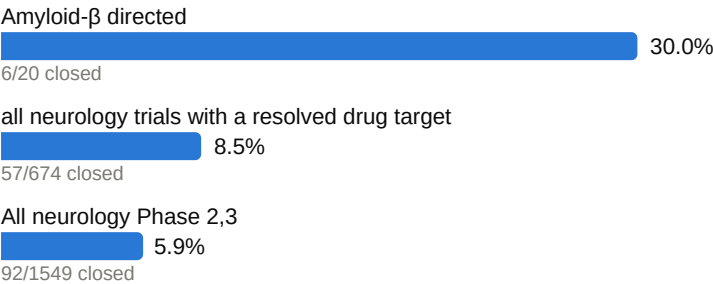
6 stopped trials, but 3 molecules — all monoclonal antibodies: Aducanumab, Crenezumab and Solanezumab. 4 of the stops were the trial's own verdict and 1 followed a decision taken elsewhere; 1 cannot be established. Against 8.5% for all neurology trials with a resolved drug target and 5.9% across all neurology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

30.0%	6 / 0 / 0	71.4%	11.0%
6 of 20 closed trials (95% CI 14.5%–51.9%)	efficacy / safety / both (adds to 6)	of 28 trials have closed · 8 still open or unresolved	stopped for a biological reason within 3 years of starting (95% CI 3.7%–30.5%; 16 still at risk; 3.2% area-wide)

1 of 3

Completed and missed the primary endpoint. Of 5 completed trials in this segment, 3 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 1 missed and 2 met. Counted separately and never in the rate above; a floor, not a rate. 1 of the 5 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 9 distinct experimental drugs across 28 trials (antibody 27, small molecule 2, unknown 1). Most tested: DONANEMAB (7), GANTENERUMAB (7), ADUCANUMAB (5), LECANEMAB (4), CRENEZUMAB (3). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 6 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (Biogen) contributed 3; removing that programme's trials from both sides leaves 20.0%. Screened alongside 27 other segments in this area, it does not survive a 10% false-discovery correction (q=0.175, Benjamini-Yekutieli). A further 9 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 75.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	5	6	83.3%	43.6%–97.0%
2018–2020	1	10	10.0%	1.8%–40.4%

Where the stops sit

Efficacy / futility: 6.
Sponsors with most stops: Biogen (3), Roche (2), Eli Lilly (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02477800	3	Biogen BIIB	ADUCANUMAB	2019-08	Efficacy	Study was discontinued based on futility analysis done and not based on safety concerns. Follow-up visits and closing out study activities are completed

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02484547	3	Biogen 	ADUCANUMAB	2019-08	Efficacy	Study was discontinued based on futility analysis done and not based on safety concerns. Follow-up visits and closing out study activities are completed
NCT03639987	2	Biogen 	ADUCANUMAB	2019-07	Efficacy	Study was discontinued based on futility analysis conducted on Phase 3 trials (NCT02477800 and NCT02484547) and not based on safety concerns.
NCT03114657	3	Roche	CRENEZUMAB	2019-06	Efficacy	This study was discontinued due to an interim analysis in the BN29552 study, which indicated that Crenezumab was unlikely to meet its primary endpoint.
NCT02670083	3	Roche	CRENEZUMAB	2019-05	Efficacy	This study was discontinued due to an interim analysis in this study, which indicated that Crenezumab was unlikely to meet its primary endpoint.
NCT02760602	3	Eli Lilly 	SOLANEZUMAB	2017-05	Efficacy	Insufficient scientific evidence that solanezumab would likely demonstrate a meaningful benefit to participants with prodromal AD as defined by study protocol.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 neurology trials started 2015–2024 that have closed (completed or terminated).
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 45% of industry neurology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 71.4% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The evidence package names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: clinicaltrialfailures.com/asset-check

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.